Lecture 03

Vector Spaces, Subspaces, Linear Independence, Basis, and Dimension

Instructor: Dr. M. Asad Iqbal, Assistant Professor
Department of Information Technology
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Lecture Focus

Understanding valid vector spaces, detecting redundancy, constructing efficient bases, and interpreting rank and dimension in machine-learning feature spaces. **[Data Eng. Link: Feature Selection, Data Compression]**

1 Lecture Overview

Data often contains repeated, dependent, or unnecessary information. Vector-space concepts provide a precise language for identifying this structure. Linear independence tells us whether features contribute genuinely new directions, while a basis gives a compact set of directions sufficient to represent the entire space.

Learning Outcomes

After completing this lecture, students should be able to:

- Explain the defining properties of a vector space.
- Determine whether a subset is a subspace.
- Distinguish between linearly independent and dependent sets.
- Test independence using a matrix and its rank.
- Define a basis and calculate coordinates relative to a basis.
- Explain dimension, rank, nullity, and the rank-nullity relation.
- Identify redundant features in a data matrix.
- Implement these concepts using Python and NumPy.

Conceptual Roadmap

Concept	Central Question
Vector space	Which objects and operations obey the linear-algebra rules?
Subspace	Which smaller set remains a vector space?
Independence	Does every vector add a genuinely new direction?
Basis	What is the smallest complete coordinate system?
Dimension	How many independent directions does the space contain?
Rank	How many independent directions are carried by a matrix?
Nullity	How many input directions are mapped to zero?

2 Vector Spaces

A vector space V over $\mathbb{R}$ is a set equipped with vector addition and scalar multiplication. These operations must behave consistently with ordinary algebra.

For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{R}$:

1. $\mathbf{u} + \mathbf{v} \in V$ and $a\mathbf{u} \in V$ (closure).
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ and $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
3. A zero vector $\mathbf{0} \in V$ exists and $\mathbf{v} + \mathbf{0} = \mathbf{v}$.
4. Every $\mathbf{v}$ has an additive inverse $-\mathbf{v}$.
5. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$.
6. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.
7. $a(b\mathbf{v}) = (ab)\mathbf{v}$ and $1\mathbf{v} = \mathbf{v}$.

Common Examples

Vector Space	Description
$\mathbb{R}^n$	All real vectors having n entries
P_n	All real polynomials of degree at most n
$M_{m\times n}$	All real $m \times n$ matrices
Function spaces	Functions satisfying specified linear conditions
Null space	Solution of $A\mathbf{x} = \mathbf{0}$

Non-Examples

The positive quadrant $\{(x, y) : x \geq 0, y \geq 0\}$ is not a vector space because multiplying a non-zero vector by -1 leaves the set. Similarly, $\{(x, y) : x + y = 1\}$ is not a vector space because it does not contain the zero vector.

[ML/DS Connection: Feature vectors usually live in $\mathbb{R}^n$. Embeddings, parameter vectors, gradients, and activation vectors are therefore elements of vector spaces, which allows ML algorithms to add, scale, compare, and transform them consistently.]

3 Subspaces

A subset W of a vector space V is a subspace if W is itself a vector space under the same operations.

3.1 Subspace Test

A non-empty subset $W \subseteq V$ is a subspace if:

1. $\mathbf{0} \in W$.
2. $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$.
3. $c \in \mathbb{R}$ and $\mathbf{u} \in W \Rightarrow c\mathbf{u} \in W$.

Important Subspaces Associated with a Matrix

For $A \in \mathbb{R}^{m\times n}$:

- The **column space** is the span of the columns of A and lies in $\mathbb{R}^m$.
- The **row space** is the span of the rows of A and lies in $\mathbb{R}^n$.
- The **null space** is $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$.

3.2 Worked Example

Consider

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}.$$

The zero vector satisfies the equation. The defining equation is homogeneous and is preserved by addition and scalar multiplication, so W is a subspace.

Let $y = s$ and $z = t$. Then $x = -2s + t$, giving

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2s + t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$W = \operatorname{span}\left(\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right).$$

[ML/DS Connection: Solution sets of homogeneous systems $A\mathbf{x} = \mathbf{0}$ are always subspaces. Solution sets of non-homogeneous systems $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ are generally not subspaces because they do not contain the zero vector.]

4 Linear Independence

Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are linearly independent if

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

has only the trivial solution $c_1 = \cdots = c_k = 0$. They are linearly dependent if a non-trivial solution exists. In that case, at least one vector can be expressed as a linear combination of the others.

4.1 Simple Examples

1. $[1, 0]^T$ and $[0, 1]^T$ are independent because neither repeats the direction of the other.
2. $[1, 2]^T$ and $[2, 4]^T$ are dependent because the second is twice the first.
3. Any set containing the zero vector is dependent.
4. More than n vectors in $\mathbb{R}^n$ must be dependent.

4.2 Matrix Rank Test

Place the vectors as columns of $A = [\mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_k]$. The vectors are independent exactly when

$$\operatorname{rank}(A) = k.$$

4.3 Polynomial Example

The polynomials $1 + x$, $x + x^2$, and $1 + 2x + x^2$ are dependent because

$$(1 + x) + (x + x^2) - (1 + 2x + x^2) = 0.$$

[ML/DS Connection: Dependent feature columns carry redundant information. Perfect dependence causes rank deficiency and can make model parameters non-unique. Near dependence, called multicollinearity, can make estimates unstable even when the matrix is technically full rank.]

5 Basis and Coordinates

A set $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_k\}$ is a basis for a vector space V when:

1. The vectors in B are linearly independent.
2. B spans V.

A basis is therefore a complete representation with no redundant directions.

5.1 Standard Basis

The standard basis of $\mathbb{R}^3$ is:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Every $[x, y, z]^T$ has the unique representation $x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$.

5.2 Coordinates Relative to a Basis

Let $B = \{\mathbf{b}_1, \mathbf{b}_2\} = \{[1, 1]^T, [1, -1]^T\}$ and $\mathbf{x} = [4, 2]^T$. We solve

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

This gives $c_1 = 3$ and $c_2 = 1$, so the coordinate vector of $\mathbf{x}$ relative to B is

$$[\mathbf{x}]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Why Independence Is Essential

If the basis vectors were dependent, a vector could have multiple coordinate descriptions. Independence guarantees uniqueness; spanning guarantees that every vector in the space can be represented.

[ML/DS Connection: Changing a basis changes coordinates without changing the underlying object. PCA later constructs a new orthogonal basis aligned with directions of greatest variation in the data.]

6 Dimension, Rank, and Nullity

The dimension of a finite-dimensional vector space is the number of vectors in any basis of that space. Different bases may contain different vectors, but they always contain the same number of vectors.

$$\dim(\mathbb{R}^n) = n, \quad \dim(P_n) = n + 1.$$

6.1 Rank

For a matrix A:

$$\operatorname{rank}(A) = \dim(\operatorname{Col}(A)) = \dim(\operatorname{Row}(A)).$$

Rank is the number of independent columns, equivalently the number of independent rows. A matrix has full column rank when every column adds an independent direction.

6.2 Nullity and Rank-Nullity

The nullity of A is the dimension of its null space. If A has n columns, then

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = n.$$

If A has four columns and rank 2, then its nullity is 2. This means two independent input directions are mapped to zero.

6.3 Data Dimension versus Intrinsic Dimension

A dataset may have 100 recorded features but lie close to a much lower-dimensional structure. The number of columns is its ambient dimension; the number of meaningful independent directions is related to its intrinsic dimension.

[ML/DS Connection: Dimensionality-reduction methods seek a smaller set of informative directions. This reduces storage and computation, supports visualization, and may reduce noise or multicollinearity.]

7 Python and NumPy Practice

7.1 Testing Linear Independence

```python
import numpy as np

def are_independent(vectors, tol=None):
    A = np.column_stack(vectors)
    rank = np.linalg.matrix_rank(A, tol=tol)
```

```python
    return rank == len(vectors)

v1 = np.array([1.0, 2.0])
v2 = np.array([2.0, 4.0])
v3 = np.array([0.0, 1.0])
print(are_independent([v1, v2]))  # False
print(are_independent([v1, v3]))  # True
```

7.2 Coordinates Relative to a Basis

```python
B = np.array([[1.0, 1.0], [1.0, -1.0]])
x = np.array([4.0, 2.0])
coordinates = np.linalg.solve(B, x)
print("Coordinates:", coordinates)
print("Reconstruction:", B @ coordinates)
```

Expected coordinates are $[3., 1.]$, and the reconstruction equals the original vector.

7.3 Rank and Nullity

```python
A = np.array([
    [1.0, 2.0, 3.0, 4.0],
    [0.0, 1.0, 1.0, 2.0],
    [1.0, 3.0, 4.0, 6.0]
])
rank = np.linalg.matrix_rank(A)
nullity = A.shape[1] - rank
print("Rank:", rank)
print("Nullity:", nullity)
```

8 ML Case Study: Redundant Features

Suppose a dataset records normalized attendance, quiz performance, and a third feature created by adding the first two. The third column contains no new information.

```python
attendance = np.array([0.80, 0.60, 0.90, 0.70])
quiz = np.array([0.70, 0.85, 0.65, 0.75])
combined = attendance + quiz
X = np.column_stack((attendance, quiz, combined))

print("Data matrix:\n", X)
print("Number of features:", X.shape[1])
print("Rank:", np.linalg.matrix_rank(X))
```

The matrix has three columns but rank 2 because $\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{x}_2$.

8.1 Selecting Independent Columns

```python
def independent_column_indices(A, tol=None):
    selected = []
```

```python
    current = np.empty((A.shape[0], 0))
    current_rank = 0

    for j in range(A.shape[1]):
        candidate = np.column_stack((current, A[:, j]))
        candidate_rank = np.linalg.matrix_rank(candidate, tol=tol)
        if candidate_rank > current_rank:
            selected.append(j)
            current = candidate
            current_rank = candidate_rank
    return selected

print("Independent column indices:", independent_column_indices(X))
```

[ML/DS Connection: Removing an exactly redundant feature does not remove information. However, feature selection in real projects should consider prediction value, interpretability, noise, fairness, and domain meaning, not rank alone.]

9 Review and Independent Practice

Practice Tasks

1. Determine whether $W = \{(x, y, z) : x - y + z = 0\}$ is a subspace of $\mathbb{R}^3$. Find a spanning set for it.
2. Determine whether $\{[1, 2]^T, [2, 1]^T\}$ is linearly independent using both an equation and matrix rank.
3. Explain why $\{[1, 0]^T, [0, 1]^T, [1, 1]^T\}$ spans $\mathbb{R}^2$ but is not a basis.
4. Find the coordinates of $[5, 1]^T$ relative to $B = \{[1, 1]^T, [1, -1]^T\}$.
5. Decide whether $1 + x$, $x + x^2$, and $1 + 2x + x^2$ are independent in P_2.
6. Create a 5×4 NumPy matrix whose fourth column is the sum of the first two. Calculate its rank and nullity.
7. Explain the difference between the recorded feature dimension and intrinsic dimension of a dataset.

Key Takeaways

- A vector space is closed under vector addition and scalar multiplication.
- A subspace contains the zero vector and remains closed under both operations.
- Independent vectors contribute distinct directions; dependent vectors contain redundancy.
- A basis is an independent spanning set and provides unique coordinates.
- Dimension counts basis directions, rank counts independent matrix directions, and nullity counts lost input directions.
- Feature matrices with dependent columns are rank deficient and may produce non-unique model parameters.

Key Terms

Vector space, Subspace, Closure, Linear independence, Linear dependence, Basis, Coordinates, Dimension, Rank, Nullity, Column space, Intrinsic dimension.